
FINAL PROJECT

In view of obtaining the State Engineer diploma

**Study of short-circuit currents
and optimization of the selectivity
of HTA/BT protections**

Prepared by:

Noureddine Qarafi

Supervised by:

M. Aissaoui Soufiane

Host organization: OCP Group - Jorf Lasfar

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At the end of this final study project, I would like to express my sincere thanks to all the people who contributed to its successful completion. I would particularly like to thank Mr. Aissaoui Soufiane for his guidance, availability, technical directions, and the discussions that allowed me to deepen my study of industrial electrical networks and HTA/BT protections. I also thank all the staff of the OCP Jorf Lasfar Fertilizer Zone for their welcome, collaboration, and the technical information provided to me.

Summary

This report presents a targeted study on the operational safety of the electrical network of the old fertilizer production lines at OCP Jorf Lasfar. The work mainly focuses on the analysis of the HTA/BT architecture, the calculation of impedances and short-circuit currents, and then on the verification and optimization of the selectivity of protections. The calculations are performed using the impedance method, in accordance with the IEC 60909 standard approach, and then compared with the simulation results under ETAP. The coordination study relies on the ANSI 50/51 overcurrent functions and on an ampere-time grading of the different relays to isolate the fault as close to its origin as possible while maintaining service continuity.

Keywords: short-circuit, impedance, IEC 60909, HTA/BT, ETAP, protection relays, selectivity, ANSI 50/51.

Work synthesis. The study is organized around four axes: presentation of the OCP industrial context and the HTA/BT network, calculation of equivalent impedances, determination of short-circuit currents according to IEC 60909, and selective coordination of protections. Validation under ETAP allows for the confrontation of analytical results with simulations, while the settings of the relays are prioritized so that the protection closest to the fault triggers first, with upstream levels maintaining a backup role.

Key results. The values retained in the reference document lead to a three-phase short-circuit current of 20.49 kA at the HTA busbar 07EM13.1 and 29.21 kA at the BT busbar 07EB41.1. The studied time coordination respects a progression from downstream to upstream; for the C08 - ABB panel - coupling chain, the main time delays are 0.05 s, 0.08 s, and 0.30 s, respectively.

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General introduction

In an industrial installation, the continuity of the power supply directly affects the availability of processes. A short-circuit fault can impose significant thermal and electrodynamic stresses on cables, busbars, transformers, and circuit breakers. Therefore, the sizing of equipment and the adjustment of protections require a reliable understanding of the maximum and minimum fault currents.

The studied network supplies several critical equipment in the Fertilizer workshop of Jorf Lasfar. The objective of this work is to characterize the HTA/BT network, to calculate its equivalent impedances and short-circuit currents at the main busbars, and then to analyze the coordination of protections in order to achieve selective elimination of faults. The section related to harmonics and energy metering is not addressed in this report.

The approach includes: the study of the OCP environment and the Fertilizer process, the analysis of HTA/BT positions, the manual calculation of short circuits, validation under ETAP, and the study of the amperometric and time-selectivity of the protections.

Presentation of OCP and project framework

1.1 Presentation of the OCP Group

Founded in 1920, the OCP Group plays a central role in the exploitation, processing, and valorization of Moroccan phosphate. Its activities cover mining, phosphoric acid production, and fertilizer manufacturing.

The Jorf Lasfar complex is one of the main industrial hubs of the group, encompassing production, storage, utility, and export units.

Element	Information
Company Name	Chérifien Office of Phosphates - OCP
Creation	Decree of 07/08/1920
Creation of OCP SA	2008
Headquarters	Casablanca
Main products	Phosphate, phosphoric acid, fertilizers
Processing sites	Jorf Lasfar, Safi

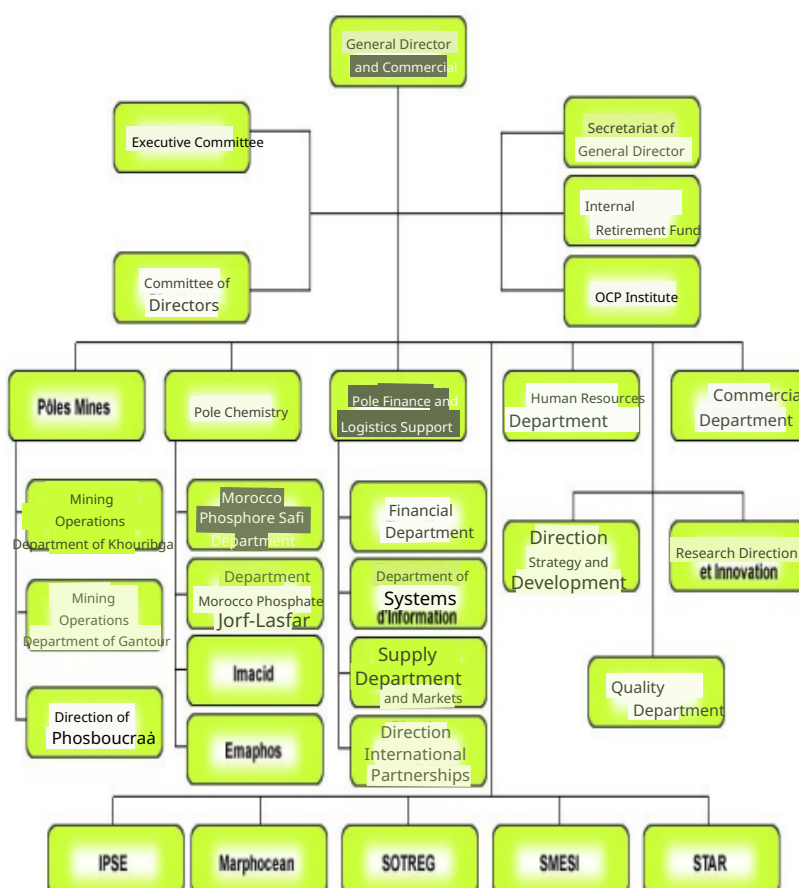


Fig. 1.1 : *Flowchart of OCP (figure taken from the source document).*

1.2 History and main activities

The evolution of the group has been gradual, from the exploitation of the first deposits in the early 20th century to the development of large chemical platforms in Safi and Jorf Lasfar. The Jorf Lasfar site started in 1986 with Maroc Phosphore, and its processing and fertilizer production capacities have been continuously strengthened.

The value chain includes the extraction and enrichment of phosphate, its transformation into phosphoric acid, and then its conversion into finished products. The production of sulfuric acid provides the necessary reagent for the attack on phosphate. The phosphoric acid obtained is then used, particularly with ammonia, for the production of various phosphate fertilizers.

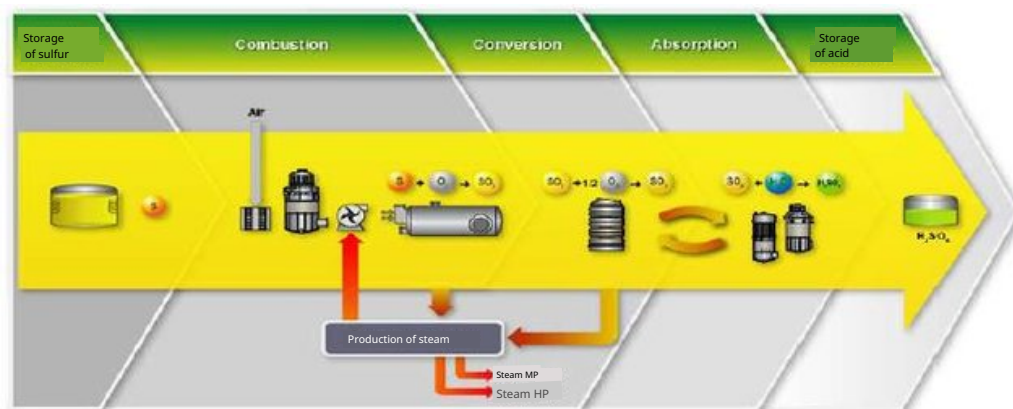


Fig. 1.2 : *Sulfuric acid production process (source document).*



Fig. 1.3 : *Illustration of fertilizers (source document).*

1.3 Jorf Lasfar Site and Fertilizer Workshop

The study is linked to the Fertilizer entity of Maroc Phosphore III and IV. The old lines 07A, 07B, 07C, and 07D were commissioned in 1986 with a capacity of approximately 90 t/h each. Three more recent lines were then added: 107A, 107B, and 107C, with a reported capacity of 120 t/h per line. The facility also includes ammonia storage units and other utilities necessary for the continuity of the process.



Fig. 1.4 : *Location of the internship site in Jorf Lasfar (source document).*



Fig. 1.5 : Location of the Fertilizer Unit (source document).

1.4 Fertilizer production process

The production of fertilizers is based on the neutralization of phosphoric acid by ammonia, followed by granulation. The product is then dried, screened, and, if necessary, crushed and recycled. After cooling, a coating limits caking and dust emission during storage. The gases from the process are dusted and washed before being released.

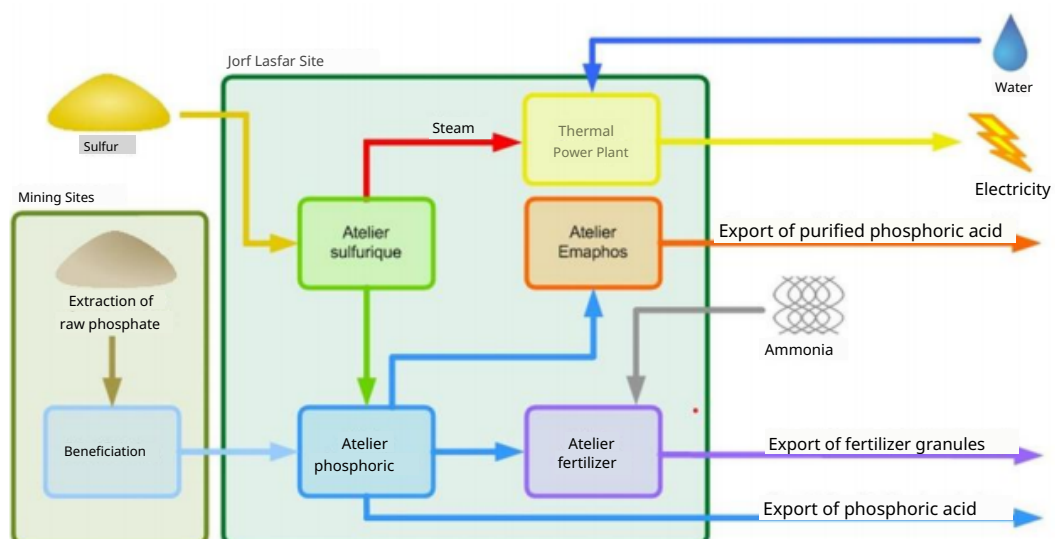


Fig. 1.6 : Connections between the workshops of the Jorf Lasfar complex (source document).

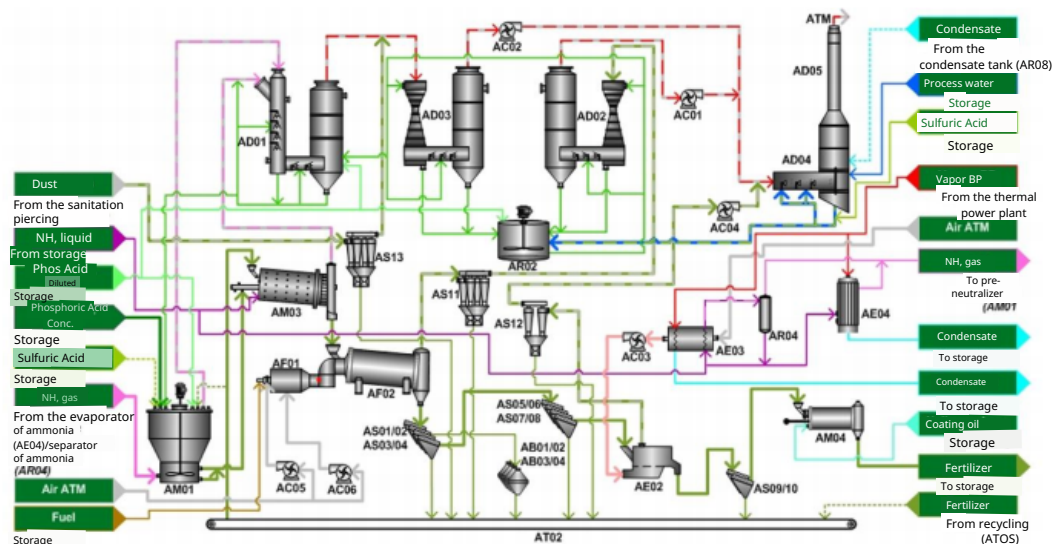


Fig. 1.7 : Overall diagram of the fertilizer production process (source document).

1.5 Context and Issues

The electrical network of the production lines must ensure a high continuity of service. The document studied notably reports an incident of short-circuit that led to the shutdown of several production lines, illustrating the industrial impact of poor fault elimination. Insufficient coordination can lead to upstream protections being triggered when a device closer to the fault should have intervened first.



Fig. 1.8 : *Diagram illustrating a short-circuit fault in the installation (source document).*

1.6 Selected Objectives

The scope of this report is limited to the following objectives: analyze the HTA/BT posts, calculate short-circuit currents using the impedance method, verify the results using ETAP, examine the protection settings, and optimize selectivity between arrivals, coupling, transformers, panels, and motors.

Analysis of the HTA/BT network and calculation of short-circuit currents

2.1 General distribution architecture

The supply of the old fertilizer lines is integrated into an interconnected industrial network. The HTA substations distribute energy to the HTA/BT transformers and large motors, while the low voltage panels supply auxiliaries and power departures. The network includes HAZE-MEYER and ABB panels, as well as 3150 kVA transformers supplying the BT busbars.

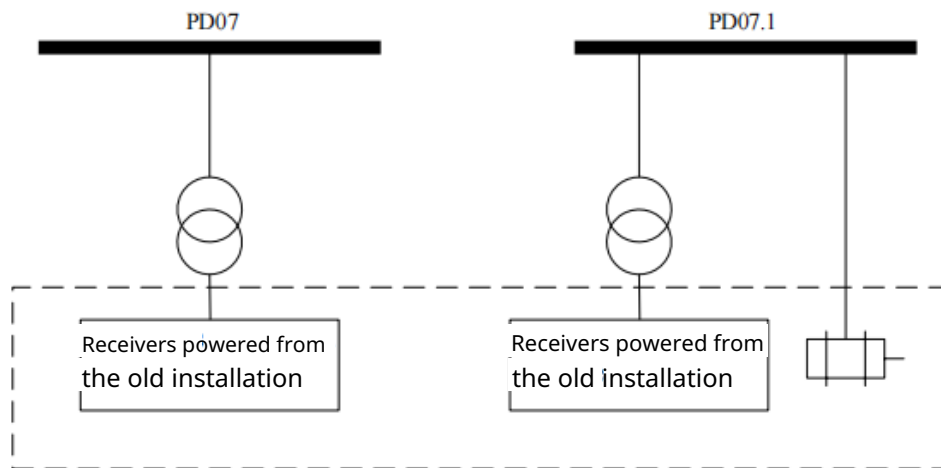


Fig. 2.1 : *Production line diagram for fertilizers (source document).*

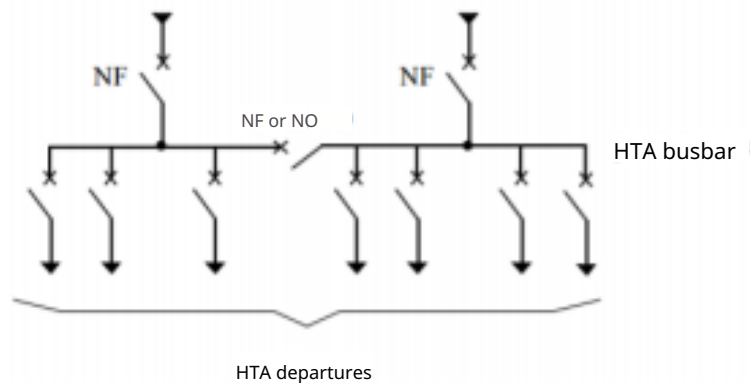


Fig. 2.2 : *Principle of double antenna busbar (source document).*

2.2 Principle of short-circuit calculation

The calculation of short-circuits is essential to verify the breaking capacity of circuit breakers, their closing capacity, and the electrodynamic withstand of equipment. Two situations must be distinguished: the maximum current, used for equipment sizing, and the minimum current, used to check the sensitivity of protections.

The chosen method is the impedance method based on the Thévenin equivalent. Each source, cable, transformer, or machine is represented by a resistance and a reactance. The components are added separately up to the fault point to obtain the equivalent impedance.

$$Z = \sqrt{R^2 + X^2}$$

This approach is compatible with IEC 60909 and with the French calculation guides for HT/BT installations. In the original PDF, the reference "UTE C 13-2015" is likely a typo; the corresponding guide for the HT domain is UTE C 13-205.

2.3 Impedance of network elements

2.3.1 Upstream network

When the short-circuit power S_{cc} of the upstream network is known, its equivalent impedance is obtained from the composite voltage U :

$$Z_a = \frac{U^2}{S_{cc}}$$

For the studied 10 kV network, the separation of the impedance into resistance and reactance is performed using the R/Z ratio adopted in the calculation document.

2.3.2 Cable connections

The resistance of a connection depends on the resistivity of the conductor, its length, and its cross-section:

$$R_L = \rho \frac{l}{S}$$

The reactance is determined from the linear inductance and the pulsation of the network:

$$X_L = \omega L$$

HTA cable section	Installation in a trefoil arrangement	Installation in a flat arrangement
240 mm ²	0,342	0.541
95 mm ²	0,399	0,625

2.3.3 Transformers

The equivalent impedance of a transformer is calculated from its short-circuit voltage U_{cc} and its rated power S_n :

$$Z_T = \frac{U_{cc}}{100} \cdot \frac{1}{S_n}^2$$

For transformers of 3150 kVA, 10 kV/695 V, with $U_{cc} = 7.13\%$, the calculation in the document leads to an equivalent impedance of approximately 0.010887 Ω .

2.3.4 Asynchronous Machines

During a fault, asynchronous motors can temporarily contribute to the short-circuit current by returning the stored electromagnetic and mechanical energy. Their contribution is represented by a subtransient reactance.

It becomes particularly significant when several high-power motors are connected near the fault point.

2.4 Modeling of Connections

The HTA substations are connected to the PJ5 substation by armored copper cables of 240 mm², with two cables per phase and an insulation voltage of 24 kV. The motors are connected by 95 mm² cables. The impedances calculated in the source document are summarized below.

Element	R (Ω)	X (Ω)	Z
HTA Post	0,030	0,080	0.030 + j0.080
Transformer - primary	5.845×10^{-3}	2.4×10^{-3}	$5.845 \times 10^{-3} + j2.4 \times 10^{-3}$
Transformer - secondary	3.37×10^{-4}	2.8×10^{-3}	$3.37 \times 10^{-4} + j2.8 \times 10^{-3}$
Moteur C02 (A,B)	0,035	0,0144	0.035 + j0.0144
Moteur C02 (C,D)	1.75×10^{-2}	7.2×10^{-3}	$1.75 \times 10^{-2} + j7.2 \times 10^{-3}$
Moteur C08	0,0467	0,0192	0.0467 + j0.0192

2.5 Three-phase short-circuit maximum at the HTA busbar

For a remote short-circuit from an alternator, the initial symmetrical three-phase current is determined according to IEC 60909 by:

$$I''_{k3} = \frac{U}{\sqrt{3} \cdot Z_{cc}}$$

At point 07EM13, the obtained short-circuit impedance is $Z_{cc} = 0.31 \Omega$. The calculation gives:

$$I''_{k3} = 20.49 \text{ kA}$$

Nominal tension One	Voltage factor c for the calculation	
	short-circuit currents maximum $c_{max}^{1)}$	of minimum short-circuit currents c_{min}
Low voltage 100 V to 1,000 V (table I of IEC 60038)	$1,05^3$ $1,10^{4)}$	0,95
Moyenne tension >1 kV to 35 kV (Table III of IEC 60038)	1,10	1,00
Haute tension²⁾ >35 kV (Table IV of IEC 60038)		

1) It is important that c does not exceed the highest voltage U_m of the equipment for power networks.

2) If no nominal voltage is defined, it is appropriate to apply
 $c_{max} U_n = U_m$ or $c\alpha = 0.90 \times U$

3) For low voltage networks, with a tolerance of +6%, for example, systems renamed from 380 V to 400 V.

4) For low voltage networks, with a tolerance of +10%.

Fig. 2.3 : Voltage factor c according to IEC 60909-0 (figure from the source document).

The peak current is calculated by:

$$i_p = \kappa \sqrt{2} I_k''$$

The factor κ depends on the R/X ratio and can be estimated by
 $\kappa = 1,02 + 0,98e^{-3R/X}$. For $R/X = 0.25$, the document obtains $\kappa \approx 1.48$, resulting in a peak current close to 42.9 kA.

2.6 Maximum short-circuit in the low-voltage busbar system

The upstream impedances must be converted to the voltage level of the fault point by the square of the transformation ratio:

$$Z_{BT} = Z_{HTA} \cdot (U_{BT} / U_{HTA})^2$$

For the busbar 07EB41.1, the equivalent impedance converted to low voltage results in a maximum symmetrical three-phase current of 29.21 kA and a peak current of 77.13 kA.

2.7 Two-phase short circuit

In the case of a two-phase fault, the relationship used in the document is:

$$I_{k2} = \frac{U}{\sqrt{2} Z_{cc}} = \frac{1}{\sqrt{2}} I_{k3}$$

We obtain 17.74 kA at busbar 07EM13.1 and 25.29 kA at busbar 07EB41.1.

2.8 Synthesis of manual results

Busbar	Ik3 (kA)	Ip (kA)	Ik2 (kA)
07EM13.1	20,49	42,97	17.74
07EB41.1	29,21	77,13	25,29

The breaking and closing powers of the considered circuit breakers remain superior to the calculated values. In low voltage, the Masterpact NW32 has breaking powers of 85 kA and 65 kA depending on the version, which are significantly higher than the maximum symmetrical current of 29.21 kA. Their closing powers are also superior to the peak current obtained.

2.9 Validation by ETAP

A modeling under ETAP was used to integrate additional parameters, including the contribution of motors and the tolerances of equipment. For the three-phase fault applied to the new HTA substation, the simulation from the source document shows a source contribution of 21.17 kA, a motor contribution of approximately 0.24 kA, and a total current at the busbars of around 21.41 kA.

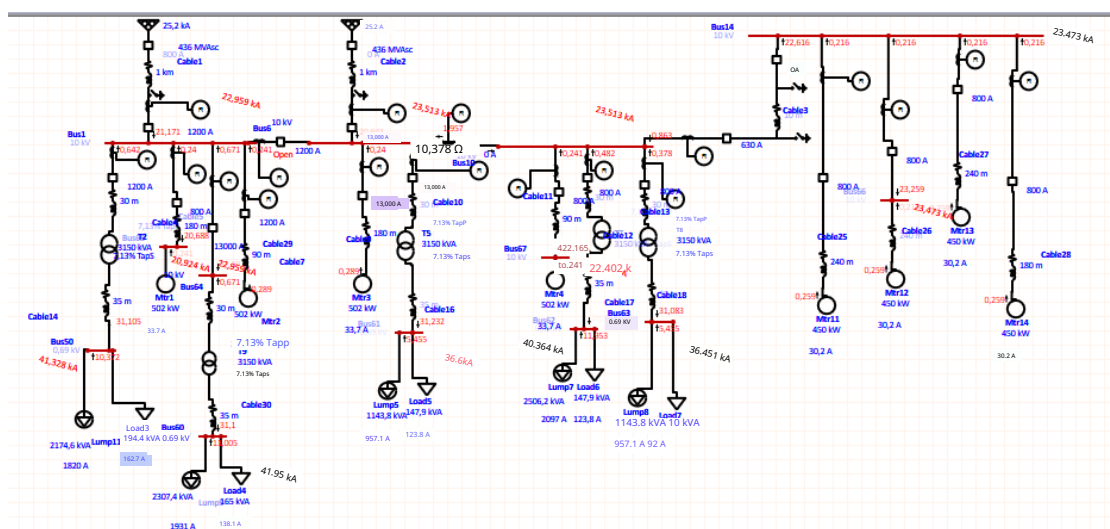


Fig. 2.4 : ETAP simulation of a three-phase fault at the new HTA substation (source document).

2.10 Minimum short-circuit currents

The minimum values mainly serve to verify that the instantaneous or time-delayed thresholds of the protections remain sufficiently sensitive. The ETAP results included in the report are as follows :

Table	Voltage (kV)	I''k2 min (kA)
07EM11.1	10,000	1.70
07EM12.1	10,000	1.69
07EM13.1	10,000	1.69
07EB41.1	0,690	9.80
07EB42.1	0,690	9.80
07EB43.1	0,690	9.79
07EB44.1	0,690	10.27
07EB47.1	0,690	9,76

Conclusion of Chapter 2

The study of impedances and short circuits has allowed for the verification of the performance of the main devices and provided the necessary values for relay adjustment. The consistency between manual calculations and ETAP simulations confirms the order of magnitude of fault currents at HTA and BT levels.

Coordination and selectivity of electrical protections

3.1 Role and qualities of a protection system

A protection system must detect an abnormal state, optimally locate the fault, and trigger the opening of the appropriate cutoff device. The desired qualities are speed, reliability, sensitivity, and above all, selectivity, which is the ability to isolate only the defective part of the network.

3.2 Protection Chain

The protection chain consists of current transformers, relays and circuit breakers. The current transformers reduce the primary currents to standardized values compatible with the inputs of the relays. The relays analyze the electrical quantities and command the opening of the circuit breaker when the set thresholds and timings are reached.



Fig. 3.1 : Protection relay used in the study (source document).

The document particularly describes the use of ABB REF615 relays on certain panels. ANSI functions 50/51 are used for three-phase faults and functions 50N/51N for single-phase faults.

3.3 Adjustment Principles

The settings must respect two contradictory constraints: they must be low enough to detect the smallest fault in the protected area, while remaining high enough not to trigger on normal currents, transient overloads, or starting current calls.

The coordination study is conducted successively on the HTA arrivals, transformer departures, coupling, the ABB panel, and motors C02 and C08.

3.4 Protection of HTA Arrivals

For arrivals 07EM13 and 07EM14, the relays have a first timed threshold $I>$ and a second threshold $I\gg$ for significant faults. The settings are chosen based on the base current of the installation and the minimum available short-circuit current.

Parameter	07EM13	07EM14
$I>$	1314.17 A	1314.17 A
$I>$	1.0 s	1.2 s
Type of curve	Defined	Inverse
$I\gg$	2359 A	2359 A
$tI\gg$	0.4 s	0.4 s

3.5 Coupling and ABB Panel Protection

The adjustment of the coupling is crucial to prevent a downstream fault from causing loss of the entire distribution. In the source document, the second threshold of the coupling is set at 1507 A. The short-circuit delay is increased from 0.05 s to 0.30 s to allow downstream protections time to eliminate the fault first.

For the ABB panel, optimization leads to a first threshold of 148.28 A and a second threshold of 1186.24 A. The fast delay of the second threshold is set at 0.08 s, placing this protection between that of motor C08 and the coupling.

3.6 Protection of Motors C02 and C08

The nominal current of a motor is estimated by:

$$I_n = P / (\sqrt{3} \cdot U \cdot \eta \cdot \cos\varphi)$$

The ANSI 51 function provides protection against overloads with a threshold above the nominal current and a delay sufficient to support starting. The ANSI 50 function ensures rapid elimination of short-circuit faults.

Parameter	C02	C08
Power	502 kW	450 kW

Parameter	C02	C08
Nominal current	33.7 A	37.6 A
I _{cc} minimal	4266 A	4627 A
I> (51)	41.36 A	37.07 A
tI>	10 s	20 s
I» (50)	300.8 A	269.6 A
tI»	0.10 s	0.05 s

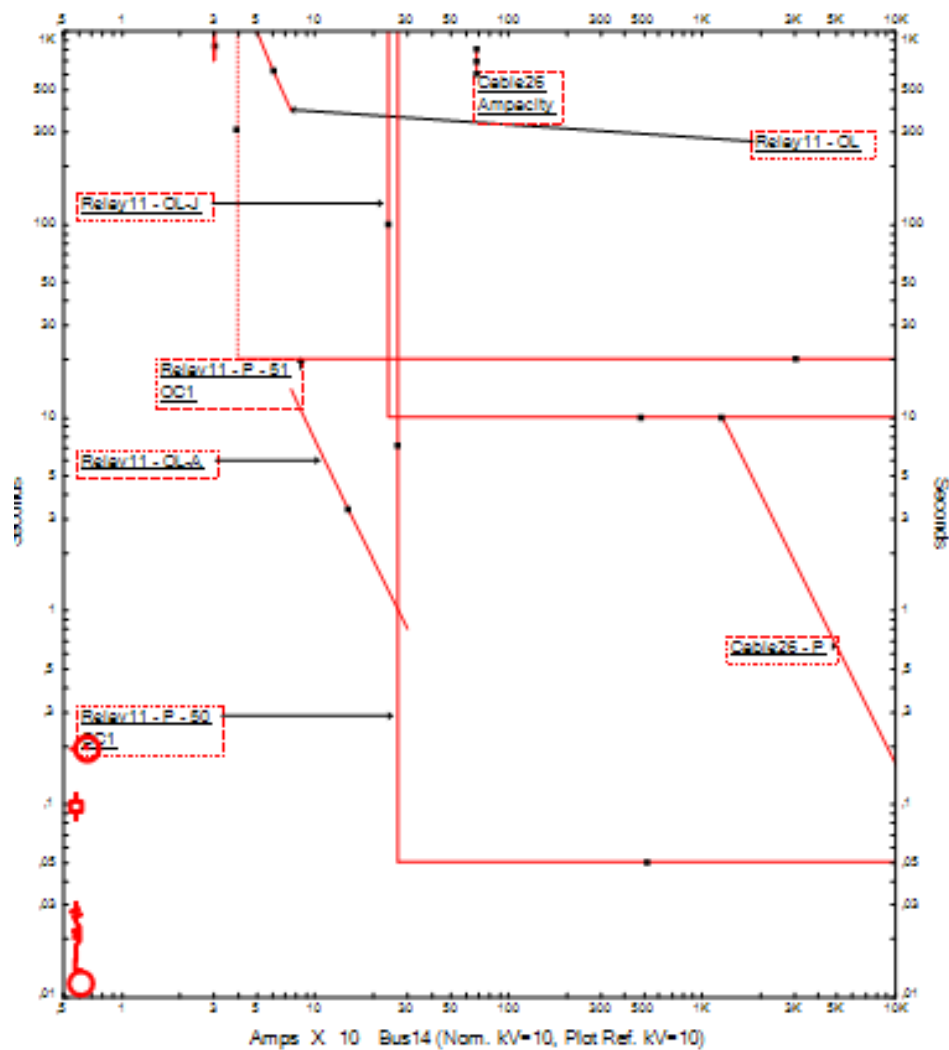


Fig. 3.2 : TCC curve of the C08 engine (source document).

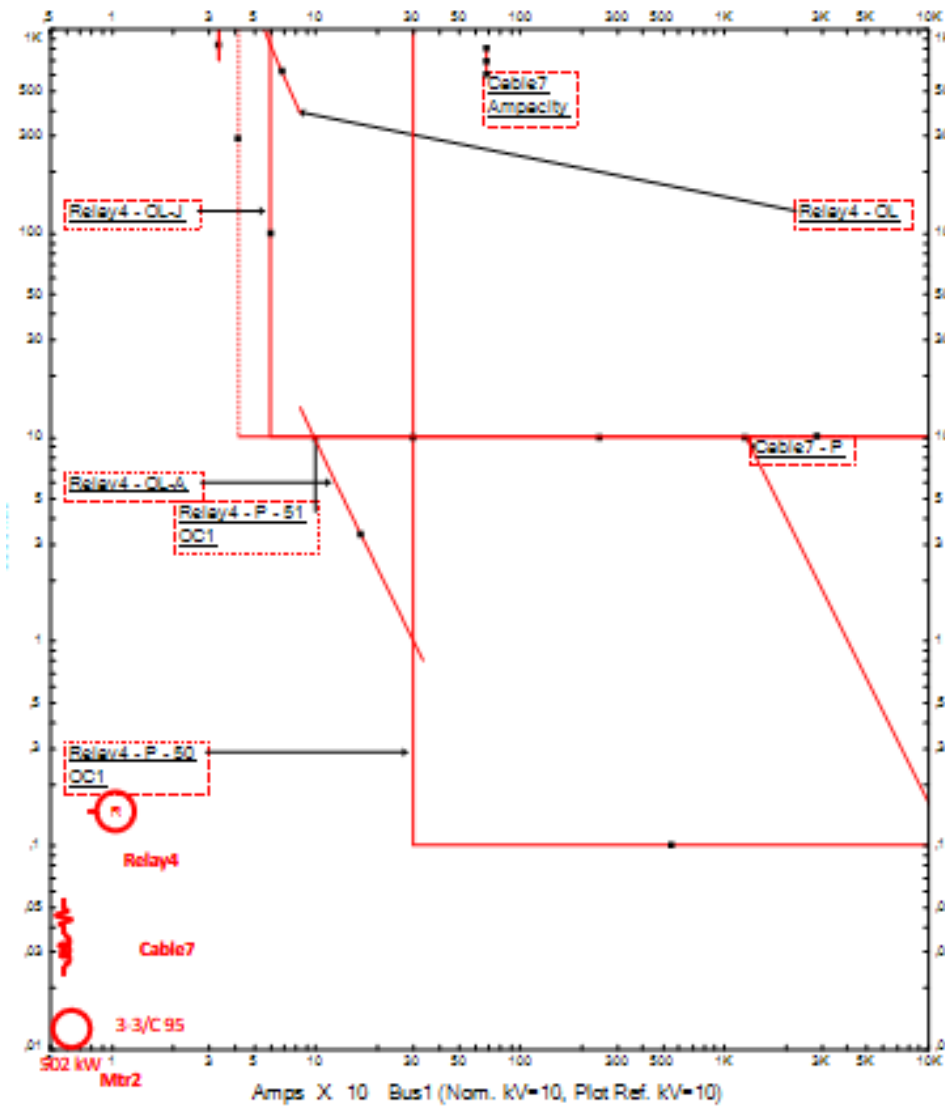


Fig. 3.3 : TCC curve of motor C02 (source document).

3.7 Study of selectivity

Selectivity is total when, for any fault in the considered range, only the protection located immediately upstream of the fault triggers. It is partial when this condition is only met up to a certain current value.

3.7.1 Ampere selectivity

Ampere selectivity exploits the variation of short-circuit current with the position of the fault. The threshold of the protection in an area must be lower than the minimum fault current in that area, but higher than the maximum current that the protection can see for a fault located further downstream. The document globally verifies the condition $I_{amont} > I_{aval}$ for the different levels.

3.7.2 Time Selectivity

Time selectivity complements current discrimination. Series protections receive increasing time delays from downstream to upstream. The rule used is:

$$t_{\text{upstream}} = t_{\text{downstream}} + \Delta t_{\text{set}}$$

$$\Delta t_{\text{set}} \approx 0,2 \text{ to } 0,3, s$$

This margin takes into account the operating time of the relay, the opening time of the circuit breaker, and timing uncertainties.

3.8 Relay Coordination

3.8.1 HTA Arrivals

For the chain including motor and transformer departures followed by the HTA arrival, the upstream settings are intentionally more delayed. Based on the most unfavorable downstream delays, a stagger of about 0.2 to 0.3 s helps avoid simultaneous tripping.

3.8.2 Coupling

After optimization, the chosen time hierarchy is:

$$t_{\text{aval}} < t_{\text{couplage}} < t_{\text{amont}}$$

Motor/transformer departures eliminate local faults first; the coupling occurs at the second level with $t \approx 0.30$ s; the HTA arrival provides backup with a longer delay.

3.8.3 Motor C08 - ABB Panel - Coupling

The fastest selectivity chain is organized as follows:

$$t_{C08} = 0,05, s < t_{ABB} = 0,08, s < t_{\text{coupling}} = 0,30, s$$

Thus, a fault on the C08 departure must be eliminated by its own protection. The ABB panel only intervenes as backup, followed by the coupling if the downstream protections do not trigger.

3.9 Summary of Settings

Parameter	07EM13	07EM14	Transformer C02		Coupling ABB		C08
I>	1314,17	1314,17	181,9	41,36	589,82	148,28	37,07

Parameter 07EM13 07EM14 Transformer C02					Coupling ABB		C08
tI> (s)	1	1,2	10	10	1,2	10	20
I»	2359	2359	1053	300,8	1507	1186,24	269,6
tI» (s)	0,4	0,4	0,15	0,1	0,3	0,08	0,05

Conclusion of Chapter 3

The obtained coordination combines amperometric selectivity and timing selectivity. The thresholds are prioritized according to the expected levels of fault, and the delays are scaled from downstream to upstream. This organization allows for prioritizing the opening of the protection closest to the fault while maintaining backup levels.

Normative references used for calculations

4.1 IEC 60909-0 - Calculation of short-circuit currents

The main reference used for calculations is IEC 60909-0, "Short-circuit currents in three-phase AC systems - Part 0: Calculation of currents." It defines the calculation procedure for short-circuit currents in low-voltage (LV) and high-voltage (HV) three-phase networks at 50 Hz or 60 Hz. The relationships used in this report for the initial symmetrical current, the voltage factor c , and the peak current k come from this methodology.

Edition used in the source document: IEC 60909-0:2016, 2nd edition. A 3rd edition IEC 60909-0:2026 was published on July 23, 2026, and now replaces the 2016 edition. To maintain consistency with the project's calculations, the results reproduced here remain based on the 2016 edition.

Official reference: <https://webstore.iec.ch/en/publication/24100>

Current edition: <https://webstore.iec.ch/en/publication/68454>

4.2 UTE C 13-205 - HV installations

The source document mentions "UTE C 13-2015" in the impedance method section. The technical reference corresponding to the calculation of short-circuit currents in high-voltage installations is UTE C 13-205, "High Voltage Electrical Installations - Practical Guide - Determination of Conductor Sections and Selection of Protection Devices." Its Chapter C is dedicated to short-circuit currents and the values of different impedances.

Official AFNOR reference: <https://www.boutique.afnor.org/fr-fr/norme/ute-c13205/installations-electriques-a-haute-tension-guide-pratique-determination-des-fa038690/4080>

4.3 NF C 13-100 - HV delivery stations

The adjustment rules cited in the protection chapter refer to NF C 13- 100, a French standard related to delivery stations powered by a public distribution network up to 33 kV. It includes provisions related to short-circuit withstand and the determination of high-voltage protections.

Official AFNOR reference: <https://norminfo.afnor.org/norme/nf-c13-100/postes-de-livraison-alimentes-par-un-reseau-public-de-distribution-hta-jusqua-33-kv/105749>

4.4 Access to Normative Texts

The complete IEC and AFNOR texts are protected by copyright and are generally accessible via the official paid platforms of IEC and AFNOR.

The links above correspond to the official sheets of the original documents. They allow for the accurate identification of the editions and the scope of application of the standards used.

General conclusion

This work has allowed for the study of the protection part of the HTA/BT electrical network of the former fertilizer production lines of OCP Jorf Lasfar. The impedance method has been applied to determine the maximum and minimum short-circuit currents at the main voltage levels. The manual results were then compared to an ETAP modeling, with consistent values suitable for verifying the breaking and closing capacities of the circuit breakers.

The study of the settings showed the importance of a hierarchical coordination of protections. The combination of amperometric and time selectivities allows for limiting the extent of outages and maintaining a backup level in case of a failure of a downstream protection. The optimized delays of the C08 motor, the ABB panel, and the coupling illustrate this logic, with a progressive triggering of 0.05 s, 0.08 s, and then 0.30 s.

Finally, the verification of the normative references confirms that the short-circuit calculations are based on IEC 60909-0 and that the HT guide mentioned in the source text should be identified as UTE C 13-205. This clarification improves the technical traceability of the study.

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Main data from the reference document

The context figures, diagrams, and TCC curves included in this report are taken directly from the provided PDF, in accordance with the request to retain the graphical elements when no modifications were necessary. The numerical calculations and settings presented are derived from Chapters 2 and 3 of the reference document, deliberately excluding the entire section dedicated to harmonics.